Norms

1) $\|\vec{x}\| \geq 0$

2) Triangle: $\|\vec{x}+\vec{y}\| \leq \|\vec{x}\|+\|\vec{y}\|$

3) Homogeneity: $\|\alpha\vec{x}\| = |\alpha|\,\|\vec{x}\|$

ℓ_p-norm: $\|\vec{x}\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{\frac{1}{p}}$

$p=2$: $\|\vec{x}\|_2 = \sqrt{\sum x_i}$

$p=1$: $\|\vec{x}\|_1 = \sum |x_i|$

$p=\infty$: $\max_{x_1 \dots x_n} |x_i|$

Projections: ① Vector projection: $(\vec{a}^T\vec{b})\vec{b}$

② Scalar projection: $(\vec{a}^T\vec{b})$

③ Projection length: $|\vec{a}^T\vec{b}|$

($\vec{b}$ is unit norm above)

$\text{proj}_{Col(A)}\vec{y} = A(A^TA)^{-1}A^T\vec{y}$

Gram-Schmidt

Goal: given a set of vectors $\{\vec{a}_1 \dots \vec{a}_\ell\} \in \mathbb{R}^n$, find an orthonormal set of vectors $\{\vec{q}_1 \dots \vec{q}_\ell\}$ which span the same set.

for i in range(ℓ):

$\vec{z}_i = \vec{a}_i - \sum_{k=1}^{i-1} \langle \vec{a}_i, \vec{q}_k \rangle \vec{q}_k$

$\vec{q}_i = \frac{\vec{z}_i}{\|\vec{z}_i\|}$ ← discard if $\vec{z} = \vec{0}$

return $\{\vec{q}_1 \dots \vec{q}_\ell\}$

Misc Rayleigh coefficient: for $A \in S^n$, $\lambda_{min}(A) \leq \frac{\vec{x}^TA\vec{x}}{\vec{x}^T\vec{x}} \leq \lambda_{max}(A)$ $\forall x \in \mathbb{R}^n$

Spectral Theorem

$\forall A \in S^n$, $A = U\Lambda U^T$

Λ: diagonal of eigenvectors

U: orthonormal matrix where columns are eigenvectors

If $A = U\Lambda U^T$, $A^k = U\Lambda^k U^T$

Suppose $rk(A) = r$. Then, $N(A)$: last $(n-r)$ columns of U. $Col(A)$: first r columns of U.

Cauchy-Schwarz:

$\langle \vec{x}, \vec{y} \rangle = \|\vec{x}\|_2\|\vec{y}\|_2 \cos\theta \leq \|\vec{x}\|\,\|\vec{y}\|$

Inner product properties:

$\langle -2\vec{x}, \vec{y} \rangle = -2\langle \vec{x}, \vec{y} \rangle$

$\langle \vec{x}, \vec{y}+\vec{z} \rangle = \langle \vec{x}, \vec{y} \rangle + \langle \vec{x}, \vec{z} \rangle$

SVD

$$\underset{m\times n}{A} = \underset{m\times m}{U}\,\underset{m\times n}{\Sigma}\,\underset{n\times n}{V^T} = \begin{bmatrix} U_r & U_{m-r} \end{bmatrix} \begin{bmatrix} \Sigma_r & 0_{r\times(n-r)} \\ 0_{(m-r)\times r} & 0_{(m-r)\times(n-r)} \end{bmatrix} \begin{bmatrix} V_r^T \\ V_{n-r} \end{bmatrix}^T$$

U is an orthonormal matrix of eigenvectors of AA^T. $\Sigma\Sigma^T$ is the matrix of eigenvalues of AA^T

V is an orthonormal matrix of eigenvectors of A^TA. $\Sigma^T\Sigma$ is the matrix of eigenvalues of A^TA

$Col(U_r) = Col(A)$, $Col(U_{m-r}) = Null(A^T)$, $Col(V) = Col(A^T)$, $Col(V_{n-r}) = Null(A)$

$UU^T = U^TU = V^TV = U_r^TU_r = V_r^TV_r = I$ but $U_rU_r^T \neq I$ and $V_rV_r^T \neq I$

Compact SVD: $A = U_r\Sigma_rV_r^T$. Outer product SVD: $A = \sum_{i=1}^{r} \sigma_i\vec{u}_i\vec{v}_i^T$; Useful trick:

Eckart Young

① $\underset{\substack{B\in\mathbb{R}^{m\times n} \\ Rank(B)=k}}{\text{argmin}} \|A-B\|_2 = A_k = \sum_{i=1}^{k} \sigma_i\vec{u}_i\vec{v}_i^T$ ← rank k approximation

② $\underset{\substack{B\in\mathbb{R}^{m\times n} \\ Rank(B)=k}}{\text{argmin}} \|A-B\|_F = A_k$

$\|A-A_k\|_F = \|\sum_{i=k+1} \sigma_i\vec{u}_i\vec{v}_i^T\|_F = \sqrt{\sum_{i=k+1}^{n} \sigma_i^2}$

PCA

- 1st principal component is the vector that when you scalar project the points onto it, the projections have the highest variance
- Covariance matrix (for zero-mean data): $C = \frac{1}{n}X^TX = \frac{1}{n}\sum_{i=1}^{n}\vec{x}_i\vec{x}_i^T$
- 1st PC is the eigenvector corresponding to the largest eigenvalue of C
- If rows are data points, read PC's off of columns of V } SVD interpretation
- If colums are data points read PC's off of colums of U
- First PC also minimizes orthogonal distances from the points to the PC

Useful Formulas

$\vec{x}^*_{LS} = (A^TA)^{-1}A^T\vec{b}$

$\vec{x}^*_{MN} = A^T(AA^T)^{-1}\vec{b}$

$Tr(ABC) = Tr(CAB) = Tr(BCA)$

FTLA: $N(A) + Col(A^T) = \mathbb{R}^n$

Rank-nullity: # cols = dim(span(A)) + dim(N(A))

QR Decomposition

$A = QR = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix}\begin{bmatrix} R_1 \\ 0 \end{bmatrix} = Q_1R_1$

Q_1: Basis for $Col(A)$; Q_2: Basis for $Col(A^T)$; R: upper-triangular; Q: Orthonormal

$$\underset{m\times n}{\begin{bmatrix} A \end{bmatrix}} = \underset{m\times m}{\begin{bmatrix} Q \end{bmatrix}}\,\underset{m\times n}{\begin{bmatrix} R \end{bmatrix}}$$

Matrix Norms

Frobenius: $\|A\|_F = \sqrt{\sum_i\sum_j A_{ij}} = \sqrt{tr(A^TA)} = \sqrt{\sum_{i=1} \sigma_i(A)}$

Spectral: $\|A\|_2 = \max_{\|\vec{x}\|_2=1} \|A\vec{x}\|_2 = \max_{\vec{x}\neq\vec{0}} \frac{\|A\vec{x}\|_2}{\|\vec{x}\|_2} = \max_{\|\vec{x}\|_2=1} \vec{x}^TA^TA\vec{x} = \sqrt{\lambda_{max}(A^TA)} = \sigma_{max}(A)$

$A(\vec{x}+\Delta\vec{x}) = y + \Delta y$

$\frac{\|\Delta\vec{x}\|_2}{\|\vec{x}\|_2} \leq \frac{\sigma_{max}(A)}{\sigma_{min}(A)}\frac{\|\Delta\vec{y}\|_2}{\|\vec{y}\|_2}$ ← $\kappa(A)$

$\|A\vec{x}\|_p \leq \|A\|_p\|\vec{x}\|_p$, $\langle A, B \rangle = \sum_i\sum_j A_{ij}B_{ij} = tr(B^TA)$

PSD Matrices PSD ⇒ A_{ii} non negative

PSD ⇔ all eigenvalues nonnegative

PSD ⇔ $\vec{x}^TA\vec{x} \geq 0$ $\forall \vec{x} \in \mathbb{R}^n$

PSD ⇒ A is symmetric (except $\vec{x}=\vec{0}$)

Trick: show that A is invertible by showing that $A \succ 0$.

Vector Calculus

Taylor's Approx: $f(\vec{x}_0+\Delta\vec{x}) \approx \underbrace{f(\vec{x}_0) + \nabla f^T|_{\vec{x}=\vec{x}_0}\Delta\vec{x}}_{\text{First order: hyperplane}} + \frac{1}{2}(\Delta\vec{x})^T\nabla^2 f|_{\vec{x}=x_0}\Delta\vec{x}$

Main theorem: $f: \mathbb{R}^n \to \mathbb{R}$, differentiable everywhere. If $\vec{x}^*$ is an optimal sol'n of $\min_{\vec{x}\in\Omega} f(\vec{x})$, then $\nabla f = \vec{0}$

$f: \mathbb{R}^n \to \mathbb{R}$; Gradient: $\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$ Hessian: $\nabla^2 f(\vec{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$

$g(\vec{x}) = A\vec{x}$
$\nabla g(\vec{x}) = A$

Common patterns:

$g(\vec{x}) = \vec{x}^TA\vec{x}$ $\quad \nabla g(\vec{x}) = (A+A^T)\vec{x}$ $\quad \nabla^2 g(\vec{x}) = A+A^T$

$g(\vec{x}) = \|A\vec{x}-\vec{b}\|_2^2$ $\quad \nabla g(\vec{x}) = 2(A^TA\vec{x} - A^T\vec{b})$ $\quad \nabla^2 g(\vec{x}) = 2A^TA$

$g(\vec{x}) = \vec{y}^TA\vec{x}$ $\quad \nabla g(\vec{x}) = \vec{y}^TA$ $\quad \nabla^2 g(\vec{x}) = 0^{m\times n}$

$g(\vec{x}) = \|\vec{x}\|_2^2$ $\quad \nabla g(\vec{x}) = 2\vec{x}$ $\quad \nabla^2 g(\vec{x}) = 2I$

$g(\vec{x}) = \|\vec{x}\|_2$ $\quad \nabla g(\vec{x}) = \frac{\vec{x}}{\|\vec{x}\|_2}$

$g(\vec{x}) = \vec{c}^T\vec{x}$ $\quad \nabla g(\vec{x}) = \vec{c}$

$f: \mathbb{R}^n \to \mathbb{R}^m$; Jacobian: $D_g = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \text{gradient wrt } x_1 \\ \vdots \\ \text{gradient wrt } x_n \end{bmatrix}$

$g(\vec{x}) = A\vec{x}$, $D_g(\vec{x}) = A$

$g(\vec{x}) = f(\vec{x})\vec{x}$ $\quad D_g(\vec{x}) = \vec{x}(\nabla f(\vec{x}))^T + f(\vec{x})I$

$g(\vec{x}) = f(A\vec{x}+\vec{b})\vec{x}$ $\quad D_g(\vec{x}) = \vec{x}(\nabla f(A\vec{x}+\vec{b}))^TA + f(A\vec{x}+\vec{b})I$

Ridge Regression

- If A has eigenvector λ, $(A+cI)$ has eigenvector $\lambda + c$

$\min_{\vec{x}} \|A\vec{x}-\vec{b}\|_2^2 + \lambda\|\vec{x}\|_2^2 \Rightarrow \vec{x}^* = (A^TA+\lambda I)^{-1}A^T\vec{b}$

λ — how confident that I am that $\vec{x}$ is close to 0

- If f is convex and non-decreasing and g is convex $f(g(x))$ is convex
- Point-wise max of convex functions in $\vec{x}$ is convex
- All norms are convex
- Applying an affine transformation to the variable in a convex function also preserves convexity
- Nonnegative weighted sum of convex functions is convex
- $h(f(x))$ is convex if h is increasing and $f(x)$ is convex

Convexity

convex / not convex

C_1, C_2 are convex $\Rightarrow C = C_1 \cap C_2$ are convex

C is convex $\Leftrightarrow \forall \vec{x}_1, \vec{x}_2 \in C, \forall \theta \in [0,1], \theta\vec{x}_1 + (1-\theta)\vec{x}_2 \in C$

- Subspace of $\mathbb{R}^n$, hyperplane $\{\vec{x} \mid \vec{a}^T\vec{x} = b\}$, halfplane $\{\vec{x} \mid \vec{x}^T\vec{x} \le b\}$ all convex sets
- Applying an affine operator to a convex set results in a convex set

Convex functions

$\mathrm{dom}(f)$ must be convex $\quad \forall \vec{x}, \vec{y} \in \mathrm{dom}(f)$

Zero order condition: $\forall \theta \in [0,1]$ $f(\theta\vec{x} + (1-\theta)\vec{y}) \le \theta f(\vec{x}) + (1-\theta)f(\vec{y})$

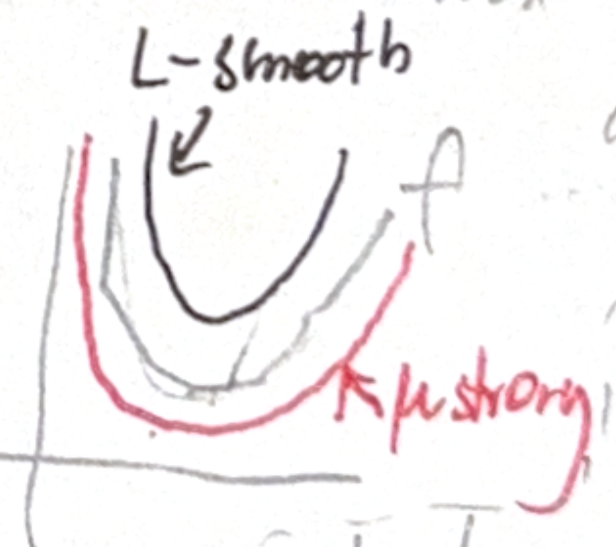
1st order condition: $f(\vec{y}) \ge f(\vec{x}) + \nabla f(\vec{x})^T(\vec{y}-\vec{x})$

2nd order condition: $\nabla^2 f(\vec{x}) \succeq 0 \quad \forall \vec{x} \in \mathrm{dom}(f)$

A function is strictly convex if the equalities above are strict

μ-strong convexity: $f(\vec{y}) \ge f(\vec{x}) + \nabla f(\vec{x})^T(\vec{y}-\vec{x}) + \frac{\mu}{2}\|\vec{y}-\vec{x}\|^2$

L-smoothness: $f(\vec{y}) \le f(\vec{x}) + \nabla f(\vec{x})^T(\vec{y}-\vec{x}) + \frac{L}{2}\|\vec{y}-\vec{x}\|_2^2$



$\mathrm{rk}(A+B) \le \mathrm{rk}(A) + \mathrm{rk}(B)$ for $A \in S^n$

Distinct eigenvalues $\to \perp$ eigenspaces

Triangle ineq. $|a+b| \le |a| + |b|$

Gradient Descent

Algorithm: $\vec{x}_{k+1} = \vec{x}_k - \eta\nabla f(\vec{x}_k)$

To prove convergence: rearrange into $\vec{x}_{k+1} - \vec{x}^* = (I - \eta\,\cdot)(\vec{x}_k - x_*)$

Then, show that this guy's eigenvalues are all $\in (-1,1)$

- An L-smooth, μ-strongly convex function converges for $\eta = \frac{1}{L}$

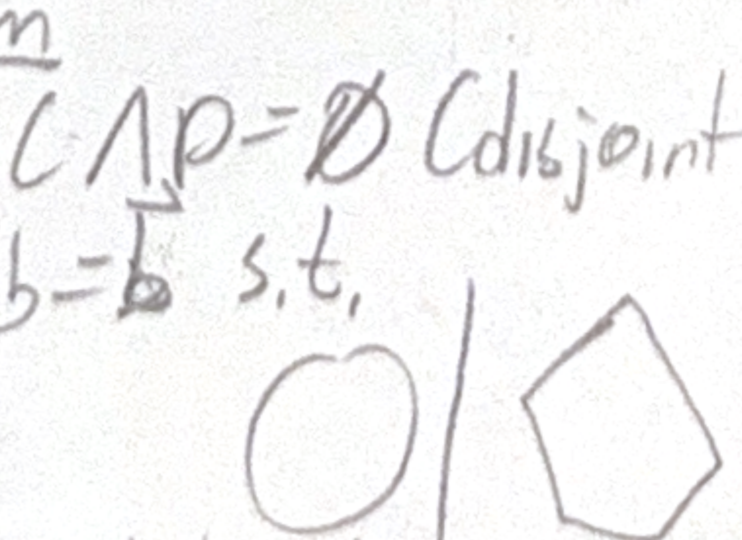
Separating Hyperplane Theorem

Let C, D be convex sets, $C \cap D = \emptyset$ (disjoint sets)

Then, $\exists$ a hyperplane $\vec{a}^Tb = \vec{b}$ s.t.

$\forall \vec{x} \in C, \ \vec{a}^T\vec{x} \ge \vec{b}$

$\forall \vec{x} \in D, \ \vec{a}^T\vec{x} \le \vec{b}$



Epigraph

$\{(x,t) \mid x \in \mathrm{dom}(f), f(x) \le t\}$

"Area above the graph"

f is convex $\Leftrightarrow \mathrm{epi}(f)$ is convex

Rayleigh Coefficient

$\lambda_{min}(A) \le \frac{\vec{x}^TA\vec{x}}{\vec{x}^T\vec{x}} \le \lambda_{max}(A), \ \forall \vec{x} \ne \vec{0}, A \in S^n$

Stochastic Gradient Descent

$$f(\vec{x}) = \frac{1}{m}\sum_{i=1}^{n} f_i(\vec{x})$$

Idea: instead of taking ∇f, choose random i and calculate ∇f_i to run GD. Use decaying step size η.

Optimization

$$p^* = \min_{\vec{x}} f_0(\vec{x})$$
s.t. $f_i(\vec{x}) \le 0 \quad i \in [m]$
$h_i(\vec{x}) = 0 \quad i \in [p]$

f_0, f_i's must be cvx, h_i's must be affine for the problem to be convex

$$L(\vec{x}, \vec{\lambda}, \vec{\nu}) = f_0(\vec{x}) + \sum_{i=1}^{m} \lambda_i f_i(\vec{x}) + \sum_{i=1}^{p} \nu_i h_i(\vec{x})$$

$$g(\vec{\lambda}, \vec{\nu}) = \min_{\vec{x}} L(\vec{x}, \vec{\lambda}, \vec{\nu}) \le p^*$$

$$d^* = \max_{\substack{\vec{\lambda} \ge 0 \\ \vec{\nu}}} g(\vec{\lambda}, \vec{\nu}) \qquad d^* \le p^* \text{ (weak duality)}$$

$p^* - d^*$ = duality gap

Minimax

$$\min_{\vec{x} \in \mathcal{X}} \max_{\vec{y} \in Y} f(\vec{x}, \vec{y}) \ge \max_{\vec{y} \in Y} \min_{\vec{x} \in \mathcal{X}} f(\vec{x}, \vec{y})$$

$$\Rightarrow p^* = \min_{\vec{x} \in \mathcal{X}} \max_{\substack{\vec{\lambda} \ge 0 \\ \vec{\nu}}} L(\vec{x}, \vec{\lambda}, \vec{\nu}) \ge d^* = \max_{\substack{\vec{\lambda} \ge 0 \\ \vec{\nu}}} L(\vec{x}, \vec{\lambda}, \vec{\nu})$$

LPs

- Feasible region forms a polytope. Optimum located at a vertex.

$$\min_{\vec{x}} \vec{c}^T \vec{x} + d \xrightarrow{\text{dual}} \max_{\vec{\lambda} \ge 0} -\vec{b}^T \vec{\lambda}$$
s.t. $A\vec{x} = \vec{b}$, $\vec{x} \ge 0$ $\qquad$ $A^T \vec{\lambda} + \vec{c} = \vec{0}$

Another common LP form: $\min_{\vec{x}} \vec{c}^T \vec{x} + d$ s.t. $A_{eq}\vec{x} = \vec{b}_{eq}$, $A\vec{x} \le \vec{b}$

QPs

convex if $H \succeq 0$

$$p^* = \min_{\vec{x}} \frac{1}{2}\vec{x}^T H \vec{x} + \vec{c}^T \vec{x} + d$$
s.t. $A\vec{x} \le b$, $C\vec{x} = \vec{d}$

For an unconstrained QP:

1. H has at least 1 eigenvalue $\Rightarrow p^* = -\infty$ (choose evec. corresponding to $-\infty$)
2. $H \succ 0$, $\vec{c} \in \mathrm{Col}(H) \Rightarrow \vec{x}^* = -H^{-1}\vec{c}$
3. $H \succeq 0$, not invertible, $\vec{c} \in \mathrm{Col}(H) \Rightarrow \vec{x}^* = H^{+}\vec{c} + \vec{\xi}$ $\qquad$ $\xi \in N(H)$ $\qquad$ $H^{+} = V\begin{bmatrix} \Sigma^{-1} & 0 \\ 0 & 0 \end{bmatrix} U$
4. $\vec{c} \notin R(H)$, $H \succeq 0 \Rightarrow p^* = -\infty$

SOCPs

[always cvx, if feasible $p^* = d^*$]

$$\min_{\vec{x}} \vec{c}^T \vec{x}$$
s.t. $\|A_i\vec{x} + \vec{b}_i\|_2 \le \vec{c}_i^T \vec{x} + \vec{d} \quad i \in [m]$

LASSO (L1) Regression

(induces sparse solutions)

$$\min_{\vec{x}} \|A\vec{x} - \vec{y}\|_2^2 + \lambda\|\vec{x}\|_1, \quad \lambda \ge 0;$$

λ parametrizes how much we care about $\|\vec{x}\|_1$ being small

Projected GD

$$\min_{\vec{x} \in \mathcal{X}} f(\vec{x})$$
$\mathcal{X}$ compact + convex set

$$\vec{x}_{k+1} = \Pi_{\mathcal{X}}(\vec{x}_k - \eta \nabla f(\vec{x}_k))$$

$$\Pi_{\mathcal{X}}(\vec{y}) = \operatorname*{argmin}_{\vec{z} \in \mathcal{X}} \|\vec{y} - \vec{z}\|_2^2$$
Π: projector

Idea: if a step takes me out of $\mathcal{X}$, project back in.

Conditional/Frank-Wolfe GD

$$\vec{y}_k = \operatorname*{argmin}_{\vec{y} \in \mathcal{X}} \nabla f(\vec{x}_k)^T \vec{y}$$

$$\vec{x}_{k+1} = (1-\delta_k)\vec{x}_k + \delta_k \vec{y}_k$$

Idea: if a step takes me outside of $\mathcal{X}$, choose vector which when dotted with the gradient is maximally negative

Slater's Condition

(to show strong duality, i.e. $d^* = p^*$)

If $\exists\, \vec{x} \in$ the relative interior of the feasible set such that $f_i(\vec{x}) < 0$ and the problem is convex, then strong duality holds

Defined Slater's: affine inequality constraints don't have to be satisfied strictly

KKT

$\tilde{x}, \tilde{\lambda}, \tilde{\nu}$ satisfy KKT conditions if

$f_i(\tilde{x}) \le 0$, $h_i(\tilde{x}) = 0$] primal feasibility

$\tilde{\lambda}_i \ge 0$] dual feasibility

$\tilde{\lambda}_i f_i(\tilde{x}) = 0$] comp. slackness

$\lambda_i > 0 \Rightarrow$ constraint i satisfied w/ equality

$\lambda_i = 0 \Rightarrow$ constraint i satisfied w/ strict inequality

$$\nabla f_0(\tilde{x}) + \sum_{i=1}^{n} \tilde{\lambda}_i \nabla f_i(\tilde{x}) + \sum_{i=1}^{p} \tilde{\nu}_i \nabla h_i(\tilde{x}) = 0$$

KKT Implications

$(\tilde{x}, \tilde{\lambda}, \tilde{\nu})$ satisfy KKT + cvx problem + h_i affine + Slater's $\Leftrightarrow (\tilde{x}, \tilde{\lambda}, \tilde{\nu})$ are primal + dual optimal

SD + $L(\tilde{x}) = g(\tilde{\lambda}, \tilde{\nu}) \Rightarrow (\tilde{x}, \tilde{\lambda}, \tilde{\nu})$ are KKT

Optimality + SD $\Rightarrow$ KKT

KKT + convex $\Rightarrow$ optimality + SD

Convexity + SD $\Rightarrow$ (KKT $\Leftrightarrow$ optimum)

Newton's Method

IDEA: descent method for cvx objective. Approximate as quadratic (Taylor approx.), find the minimizer, and repeat.

$$\vec{x}_{k+1} = \vec{x}_k - \big(\underbrace{\nabla^2 f(\vec{x}_k)}_{\text{must be PD}}\big)^{-1} \nabla f(\vec{x}_k)$$

QLQPs [cvx if H_0, H_i are PSD, H_j are 0]

$p^* = \min_{\vec x} \vec x^T H_0 \vec x + 2\vec c^T \vec x + d$

s.t. $\vec x^T H_i \vec x + 2\vec c_i^T \vec x + d_i \le 0, \ i \in [m]$

$\vec x^T H_j \vec x + 2 c_j^T \vec x + d_j = 0, \ j \in [q]$

Coordinate Descent [update coordinate by coordinate]

Initial guess: $\vec x(0) = [x_1(0) \dots x_n(0)]^T$

$x_1(k) = \arg\min_y f(y, x_2(k-1) \dots x_n(k-1))$

$x_2(k) = \arg\min_y f(x_1(k), y, x_3(k-1) \dots x_n(k-1))$

⋮

$x_n(k) = \arg\min_y f(x_1(k), x_2(k) \dots x_{n-1}(k), y)$

LQR (Linear Quadratic Regulator)

$$\min_{\substack{\vec x_1 \dots \vec x_n \\ \vec w_1 \dots \vec w_n}} \sum_{t=0}^{N-1} \frac12 (\vec x_t^T Q_t \vec x_t + \vec w^T R \vec w) + \frac12 \vec x_N^T Q_N \vec x_N$$

s.t. $\vec x_{t+1} = A\vec x_t + B\vec w_t, \ t = 0, 1 \dots N-1$

Solve w/ backwards induction, write out KKT conditions, solve for λ_N first then iterate backwards. With this new information, you can solve iteratively forwards for $\vec x$'s and $\vec w$'s.

SVMs (Support Vector Machines)

Intuition: find separating hyperplane that maximizes margin

Hard margin:

$\min_{\vec w, b} \|\vec w\|_2^2$

s.t. $y_i(\vec w^T \vec x_i + b) \ge 1$

class of training point $\in \{-1, 1\}$ — Region of

Soft margin:

$\min_{\vec w, b, \xi_i} \|\vec w\|_2^2 + C \sum_{i=1}^{n} \xi_i$ — allowing slack for points violating the margin

s.t. $y_i(\vec w^T \vec x_i + b) \ge 1 - \xi_i$

$\xi_i \ge 0$

- $\xi_i = 0 \Rightarrow$ point i is on or obeying the margin
- To make SVM into a nonlinear classifier, change kernel

$f(\vec x) = \vec w^T \vec x + b \longrightarrow f(\vec x) = \vec v^T \phi(\vec x) + b$

Equality-Constrained Newton's solving $\min_{\vec x} f(\vec x)$ s.t. $A\vec x = b$

$\Rightarrow \min \hat f(\vec x_0 + \vec v) = f(\vec x_0) + \nabla f(\vec x_0)^T \vec v + \frac12 \vec v^T \nabla^2 f(\vec x_0) \vec v$

s.t. $A(\vec x_0 + \vec v) = \vec b$

$A\vec v = 0$

step

Solve: $\begin{bmatrix} \nabla^2 f(\vec x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \vec v \\ \vec \nu \end{bmatrix} = \begin{bmatrix} -\nabla f(\vec x_0)^T \\ 0 \end{bmatrix}$

Minimax Proof

$L(\vec y) = \min_{\vec x} f_0(\vec x, \vec y) \le f_0(\vec x, \vec y) \le U(\vec x) = \max_{\vec y} f_0(\vec x, \vec y)$

$p^* = \min_{\vec x} U(\vec x) \ge L(\vec y)$

$p^* \ge \max_{\vec y} L(\vec y) = d^* \Rightarrow \min_{\vec x} \max_{\vec y} f(\vec x, \vec y) \ge \max_{\vec y} \min_{\vec x} f(\vec x, \vec y)$

Dual of QP $\succ 0$

$\min_{\vec x} \frac12 \vec x^T Q \vec x$ s.t. $A\vec x \le \vec b$

$L(\vec x, \vec\lambda) = \frac12 \vec x^T Q \vec x + \vec\lambda^T (A\vec x - \vec b)$, set $\nabla L = 0$

$g(\vec\lambda) = \frac12 \vec\lambda^T A Q^{-1} A^T \vec\lambda - \vec\lambda^T \vec b$

Dual of L1 Norm

Useful eqs: $\|\vec z\|_2 = \max_{\|\vec w\|_2 \le 1} \vec w^T \vec z$, $\|\vec z\|_\infty = \max_{\|\vec w\|_1 \le 1} \vec w^T \vec z$, $\|\vec z\|_1 = \max_{\|\vec w\|_\infty \le 1} \vec w^T \vec z$

$p^* = \min_{\vec x} \|A\vec x - \vec y\|_1$

$= \min_{\vec x} \max_{\|\vec w\|_\infty \le 1} \vec w^T (A\vec x - \vec y)$

$\ge \max \min \vec w^T (A\vec x - \vec y) = \begin{cases} -\vec w^T \vec y, & A^T \vec w = \vec 0 \\ -\infty & \text{o.w.} \end{cases}$

$\Rightarrow d^* = \max_{\vec w} -\vec w^T \vec y$ s.t. $A^T \vec w = \vec 0, \ \|\vec w\|_\infty \le 1$

Sphere Enclosure

Minimize radius of ball that encompasses all other balls with centers $\vec x_i$ and radii ρ_i.

$\min_{\vec c, r} r$ s.t. $\|\vec x_i - \vec c\|_2 + \rho_i \le r \ \ i \in [m]$

Dual of Dual of LP

PD $\Rightarrow$ Symmetric $(A^{-1})^T = (A^T)^{-1}$

$\min_{\vec x} \vec c^T \vec x$ s.t. $A\vec x = \vec b$, $\vec x \ge 0$

$L = \vec c^T \vec x + \vec\nu^T (A\vec x - \vec b) - \vec\lambda^T \vec x$

$\Rightarrow$ Dual: $\max -\vec b^T \vec\nu$

s.t. $\vec c + A^T \vec\nu - \vec\lambda = 0$, $\lambda \ge 0$ — combine constraints

$\vec c + A^T \vec\nu \ge \vec 0$

$\Rightarrow$ Dual of dual

$\max_{\vec z} -\vec c^T \vec z$ s.t. $A\vec z = \vec b$, $\vec z \ge \vec 0$